

VARUN SINGH

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SKILLS

Languages: Golang, Python, TypeScript, C++

AI / LLM: Agent Orchestration, RAG, Hybrid Search (BM25 + Vector), RRF, pgvector/HNSW, Retrieval Evaluation

Backend & Systems: FastAPI, Fastify, NestJS, PostgreSQL, Redis, gRPC, RabbitMQ, Bull, Prometheus

Infrastructure: Docker, Kubernetes, GitLab CI/CD, Artifactory, S3, Cloudflare Workers, Helm

WORK EXPERIENCE

Associate Software Developer (SDE-1)

Jul 2025 – Present

Oracle

Hybrid

- Architected Composer, an AI workflow platform that turns 5G requirements into structured test plans and automation-ready bundles; built intent-driven selection over 65k+ JSON examples, versioned generation skills, deterministic validation, resumable finite-state workflows, immutable artifacts, and approval gates, reducing manual authoring effort by 50% across 6+ network-function teams.
- Re-engineered Prometheus telemetry collection with packed PromQL requests, reusable HTTP sessions, retry handling, and bounded dual-worker concurrency, reducing end-to-end data retrieval from 8 hours to 1.5 hours; a 23.9k-request benchmark measured 75.9% lower runtime and 99.4% fewer connection establishments.
- Led a backward-compatible migration from an overloaded log server to Artifactory, introducing reusable upload abstractions and direct bastion-to-Artifactory transfers for Robot, network-function, and OSO artifacts while preserving downstream paths; also centralized Paramiko, JumpSSH, and SFTP lifecycle cleanup to eliminate recurring SSH and file-descriptor leaks.

Project Intern

Jan 2025 – Jun 2025

Oracle

Hybrid

- Developed worker-utilization reporting for performance runs and refactored dashboard state management, cutting UI-related support tickets by 80% for 75+ engineers.

Consulting Backend Engineer

Feb 2024 – Apr 2024

Trijog

Remote

- Owned end-to-end backend delivery for an LMS with 20k+ registered and 250+ peak concurrent users; built its 7-module NestJS backend from scratch with 3-role RBAC, Metabase, async jobs, and private DigitalOcean Spaces.
- Engineered Bull + FFmpeg processing for 400+ videos with multipart uploads/5 HLS renditions, cutting readiness 30→5 min; a Cloudflare Worker enforced signed, expiring auth on every manifest/segment.

PROJECTS

Repi *Autonomous log investigation engine* ([GitHub](#) · [Live Demo](#)): Designed signature-clustered log retrieval with BM25 + pgvector/HNSW fused by RRF, cross-service temporal tools, and a persistent hand-rolled finite-state machine for resumable investigations; deterministic confidence gates constrain unsupported conclusions. Migrated embeddings to ONNX with FastEmbed, shrinking the Docker image from 5.58 GB to 1.12 GB.

Collapser *gRPC request-collapsing sidecar* ([GitHub](#)): Engineered a Go gRPC sidecar with custom inflight coordination, detached backend work, TTL caching, and panic-safe cleanup; its high-contention hot path benchmarked at 82 ns/op, zero allocations, and up to 1000:1 collapse under synthetic 10k+ concurrency. Included Prometheus instrumentation.

Fend *Fleet-wide Linux hardening tool* ([GitHub](#)): Developed an Electron + React UI driving agentless, idempotent Ansible hardening across grouped Ubuntu hosts over SSH with auditable runs.

ACHIEVEMENTS

Winner, Smart India Hackathon 2023 selected among 50,000+ submissions nationwide.

Code for GovTech (C4GT) implemented a RabbitMQ + NestJS batch-processing system for digital public infrastructure, cutting document processing time by 30%.

EDUCATION

Indian Institute of Information Technology, Jabalpur

Dec 2021 – May 2025

B.Tech, Computer Science and Engineering